Semidirect products

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Semidirect product

Suppose we are given $H, K \triangleleft G$, $H \cap K = \{e\}$, and $HK = G$ (as sets). Recall that these were the conditions required for G to be expressed as the internal direct product of H and K. Also remember that to prove that $(h, k) \mapsto hk$ is an homomorphism, we needed $h_1k_1h_2k_2 = h_1h_2k_1k_2$, i.e., we needed $\forall h \in H, k \in K, hk = kh$. And to prove that this was true, we needed to use the fact that $H, K \triangleleft G$.

But what if we have $H, K \leq G$, but we do not have $\forall h \in H, k \in K, hk = kh$? (We still have that $H \cap K = \{e\}$ and $HK = G$ (as sets)).

In the most general case, instead of $hk = kh$, we have $hk = \varphi_h(k)h$, where $\varphi_h(k) = hkh^{-1}$. But in this case we are not even guaranteed that φ_h restricts to K, i.e., it is not generally true that $\forall k \in K, \varphi_h(k) \in K$.

To simplify life, instead of considering the fully general case where $H, K \leq G$, we assume that $H \leq G, K \triangleleft G$, so that we *now* have $\forall k \in K, \varphi_h(k) \in K$.

The setup therefore now becomes $H \leq G, K \triangleleft G$, $H \cap K = \{e\}$, $HK = G$ (as sets), and $\varphi : H \to \mathrm{Aut}(K) := h \mapsto (k \mapsto hkh^{-1})$. We have that $\forall h \in H, k \in K, hk = \varphi_h(k)h$. Just as previous, we want to place a suitable group structure on the Cartesian product $K \times H$ so that $\psi : K \times H \to G := (k, h) \mapsto kh$ ascends from being a set bijection to a group isomorphism. So, we just have to define what the binary operation on $K \times H$ should be.

Now, observe that, for ψ to be an isomorphism, we need $\psi((k_1, h_1))\psi((k_2, h_2)) = \psi((k_1, h_1) \cdot (k_2, h_2)) \Rightarrow k_1h_1k_2h_2 = \psi((k_1, h_1) \cdot (k_2, h_2))$. We already have by hypothesis, that $h_1k_2 = \varphi_{h_1}(k_2)h_1$. Therefore, $\psi((k_1, h_1) \cdot (k_2, h_2)) = k_1\varphi_{h_1}(k_2)h_1h_2$. From here we try to guess what $(k_1, h_1) \cdot (k_2, h_2)$ could have been (remembering the restrictions that the "$\cdot$" is associative and respects identity and inverse in $K \times H$). The guess that works is

$$(k_1, h_1) \cdot (k_2, h_2) = \left(k_1\varphi_{h_1}(k_2), h_1h_2\right)$$

And this raises $K \times H$ to a group, and by the isomorphism ψ, we say that G is the internal semidirect product $K \rtimes_\varphi H$.

Generalizing over φ, we define the external semidirect product:

> Given groups H and K, and a homomorphism $\varphi : H \to \mathrm{Aut}(K)$, we define the group $K \rtimes_\varphi H$ over the set $K \times H$ with the operation
>
> $$(k_1, h_1) \cdot (k_2, h_2) = \left(k_1\varphi_{h_1}(k_2), h_1h_2\right)$$

Observe that when we say G is the internal semidirect product of $K \triangleleft G$ and $H \leq G$, then we are using $\varphi := h \mapsto (k \mapsto hkh^{-1})$. Therefore, for internal semidirect products, we can drop the φ and just write $K \rtimes H$.

Warning!
If the set bijection that you are using is $\psi : K \times H \to G := (k, h) \mapsto kh$, then the inner product thing described above works perfectly fine. But if you place the normal subgroup on the *right*, i.e., $\psi' : H \times K \to G := (h, k) \mapsto hk$, then the operation on $H \times K$ has to be modified

$$(h_1, k_1) \cdot (h_2, k_2) = \Big(h_1 h_2, \varphi_{h_2^{-1}}(k_1) k_2\Big).$$

and you'd have to write $H \ltimes_\varphi K$ instead.

Instead of hammering through all this hogwash, just stick to writing the normal on the left and you'd be fine.

Even more generality?
The most general case, where we just have $H, K \leq G, H \cap K = \{e\}$, and $HK = G$ (as sets) is handled by the knit product, also known as the Zappa–Szép product.

Observe that direct product, semidirect product and the knit product are "similar" in the sense that

- They have the same underlying set, namely $H \times K$.
- They have subgroups isomorphic to H and K, namely $H \times \{e_K\}$ and $\{e_H\} \times K$ respectively.
- They are attempts to make a larger group out of the two given groups H and K.

The differences lie in what hk is assumed to be equal to:
- For the direct product $H \times K$, we have $hk = kh$.
- For the semidirect product $K \rtimes_\varphi H$ (where $\varphi : H \to \mathrm{Aut}(K)$), we have a "twist" on one side, $hk = \varphi_h(k)h$.
- For the knit product of H and K, we have $\varphi : H \to \mathrm{Sym}(K), \psi : K \to \mathrm{Sym}(H)$, and $hk = \varphi_h(k)\psi_k(h)$, "twisting" on both sides.

Notation alert!
The symbol $\rtimes$ is a mashup of $\times$ and $\triangleleft$. Since the normal subgroup is written towards the pointy side of $\triangleleft$, we have to write it on the left of $\rtimes$. Therefore, we have $K \triangleleft \Big(K \rtimes_\varphi H\Big)$.

Another idea
If we have $K \triangleleft G$, then we have a surjective homomorphism $f : G \to H$ such that $\ker(f) = K$. By the first isomorphism theorem, we have $G/K \cong H$.

If we can embed H back in G (i.e., we have an injective homomorphism g from H to G) such that $f|_{g(H)} = \mathrm{id}$, then G is the internal semidirect product of K and $g(H)$. We can represent G as the split extension of K and H,

$$1 \to K \to G \to H \to 1$$

An alternative way of phrasing this is that $G \cong K \rtimes_\psi G/K$ iff the quotient map $\pi : G \to G/K$ **splits**, meaning that

$$\exists \sigma : G/K \to G \text{ s.t. } \pi \circ \sigma \cong \mathrm{id}_{G/K}$$

A nice property
For groups A, B, C, given $\varphi : B \to \mathrm{Aut}(A)$, define $\psi : B \to \mathrm{Aut}(A \times C) := \psi_b((a, c)) = (\varphi_b(a), c)$. Then we have

$$\left(A \rtimes_\varphi B\right) \times C \cong (A \times C) \rtimes_\psi B$$

If we had rather defined $\psi' : B \to \mathrm{Aut}(C \times A) := \psi_b((c,a)) = (c, \varphi_b(a))$, then we'd get

$$C \times \left(A \rtimes_\varphi B\right) \cong (C \times A) \rtimes_\psi B$$

"Internal"?

Suppose we have a group G. If I have $H, K \leq G$, then G is the internal "whatever" product of H and K iff given a group structure on $H \times K$, the set function $f : H \times K \to G := (h,k) \mapsto hk$ is an isomorphism. What group structure we choose to provide on $H \times K$ decides if we get direct, semidirect or knit product.

Suppose we have a vector space V, and we have U, W as subspaces of V. We call V the internal direct sum of U and W if the default vector space on $U \times W$ is isomorphic to V via $f : U \times W := (u,w) \mapsto u + w$.

So given an algebraic structure A, and a notion of composition of elements of A (i.e., a function $* : A \times A \to A$), if we have B and C as substructures of A, then we call A as the internal product of B and C iff we can place the same algebraic structure on $B \times C$ and the set function $f : B \times C \to A := (b,c) \mapsto b * c$ is an isomorphism.

Try again later. : The last formulation is almost categorical. Refine and polish.

Nothing is "external"

Given groups A and B, and $\varphi : B \to \mathrm{Aut}(A)$, define the external semidirect product $A \rtimes_\varphi B$.

Now, we have $A \times \{e\} \triangleleft A \rtimes_\varphi B$ and $\{e\} \times B \leq A \rtimes_\varphi B$.

As in the previous section, to show that $A \rtimes_\varphi B$ is the *internal semidirect product* of $A \times \{e\}$ and $\{e\} \times B$, we have to define a group structure on the Cartesian product $(A \times \{e\}) \times (\{e\} \times B)$. To define a group structure, we just need a way define composition of two elements of $(A \times \{e\}) \times (\{e\} \times B)$, which we get naturally from the operation on $A \rtimes_\varphi B$. Now, we just have to verify that $f : (A \times \{e\}) \times (\{e\} \times B) \to A \rtimes_\varphi B \mapsto (b,c) \mapsto bc$ is an isomorphism, which is easy to do in this case.

Therefore, we have $A \rtimes_\varphi B$ as the external semidirect product of A and B and the internal semidirect product of $A \times \{e\}$ and $\{e\} \times B$.

Also observe that had we tried to build up the internal semidirect product of $A \times \{e\}$ and $\{e\} \times B$, we would have chosen $\varphi_{\mathrm{conj}} : \{e\} \times B \to \mathrm{Aut}(A \times \{e\}) := x \mapsto (y \mapsto yxy^{-1})$, which simplifies to $\varphi_{\mathrm{conj}}((e,b)) \mapsto (\varphi_b(a), e)$, and therefore, φ_{conj} is just φ up to $A \cong A \times \{e\}$ and $B \cong \{e\} \times B$.

Ahoy, mate! Stop right there!

That is, stop unless you are actually busy or something.

Suppose you are given some group G, and are told that $|G| = pq$ with $q > p$, and p, q are both primes. You can always get $H \triangleleft G$ such that $|H| = q$. You can also get $K \leq G$ such that $|K| = p$. Since $f : H \times K \to G := (h,k) \mapsto hk$ is injective in this case, so $G = HK$ (as sets), and we have $H \cap K = \{e\}$. So, are we done classifying G as internal semidirect product of H and K? Is that the only possible G?

Well, yes and no. You are correct in saying that G is just $H \rtimes_\varphi K$, where $\varphi_k = khk^{-1}$, but that doesn't fix G up to isomorphism.

Now you start screaming: H and K are fixed up to isomorphism, we have $H \cong \mathbb{Z}_p$ and $K \cong \mathbb{Z}_q$, and we also know what $\varphi : K \to \mathrm{Aut}(H)$ is. What else is there to know?

"Aha!", I say. "Think again, do you know what φ is?" You see, we know φ only if we already know how elements of K act to conjugate H. You'd know that if you had the operation on G already, but you do *not* know that.

"Hmm…", you say, nodding gravely. "Thanks for this unwanted info-dump".